

POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	Student Social Media & Mental Health Analytics
PROJECT CATEGORY	Education Analytics Mental Health Analytics Behavior Analytics
INSTITUTE NAME	Infinity Code Nexus
TRAINER / GUIDE	Mr.Anand Kumar
CREATED BY	Ms.Adurapally Divyavani
TOOLS USED	PowerBI Dashboard ,PowerBI Desktop, Power Query, DAX And PowerBI Service
SUBMISSION DATE	21-06-2026

Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

Project Overview

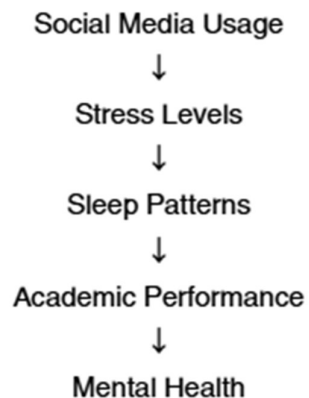
Project objectives :

Develop an interactive dashboard to analyze the impact of social media usage on student mental health.

Participants are expected to:

- ✓ Analyze student behavior
- ✓ Measure social media usage
- ✓ Evaluate stress levels
- ✓ Understand mental health patterns
- ✓ Generate actionable insights
- ✓ Present recommendations

Convert raw student data into meaningful insights that help understand the relationship between.



Business Problem

Educational institutions and student counselors want to understand:

1. How social media affects mental well-being.

A: Higher social media usage is associated with increased stress levels and can negatively impact students' mental well-being. Balanced and responsible usage helps maintain better mental health.

2. How screen time impacts academic performance.

A: Students spend more time on social media (5.08 hours/day) than studying (3.01 hours/day), indicating that excessive screen time may negatively affect academic performance and productivity.

3. How stress levels vary across different student groups.

A: High School students report the highest stress levels, followed by Undergraduate and Graduate students. This indicates that stress levels differ significantly across student groups.

4. How sleep patterns influence mental health.

A: Students with adequate sleep tend to have better mental health scores and lower stress levels, while insufficient sleep is associated with poorer mental well-being.

5. Which social media platforms have the highest usage.

A: WhatsApp is the most used social media platform, followed by TikTok and SnapChat among students.

Participants must answer:

1. What is the average mental health score?

A: Average Mental Health Score: **6.23/10**

2. What is the average social media usage per day?

A: Average Social Media Usage: **5.08** hours per day

3. Which platform is most used?

A: The platform is most used is **WhatsApp**

4. Which student groups report higher stress levels?

A: **High School** students report the highest stress levels compared to Undergraduate and Graduate students.

5. How does sleep affect mental health?

A: Effect of Sleep on Mental Health: More sleep is associated with better mental health and lower stress.

6. How does study time compare with social media usage?

A: Study Time vs Social Media Usage: Students spend more time on social media (5.08 hrs) than studying (3.01 hrs).

7. Which countries show higher mental health scores?

A: Countries with Higher Mental Health Scores: Malta, Chile, Iraq, Slovakia, and Bulgaria.

Dataset Information

Dataset Name : Student Social Media and Mental Health Impact

Item	DATASET DETAILS	Records
☐	Approximate Records	5,000+
☐	Columns	13

Project Objectives

Objective -1 : Social Media Analysis (Analyze platform usage,usage hours and engagement.)

Objective -2 : Mental Health Analysis (Evaluate mental health scores and stress levels.)

Objective -3 : Academic Analysis (Analyze study habits and academic behavior.)

Objective -4 : Lifestyle Analysis (Evaluate sleep patterns and physical activity.)

Objective -5 : Country Analysis (Compare student behavior across different countries.)

Objective -6 : Executive Reporting

Create dashboards suitable for

Schools

Universities

Counselors

Researchers

Participants should demonstrate:

✓ Data Understanding

✓ Data Analysis

✓ KPI Development

✓ Dashboard Design

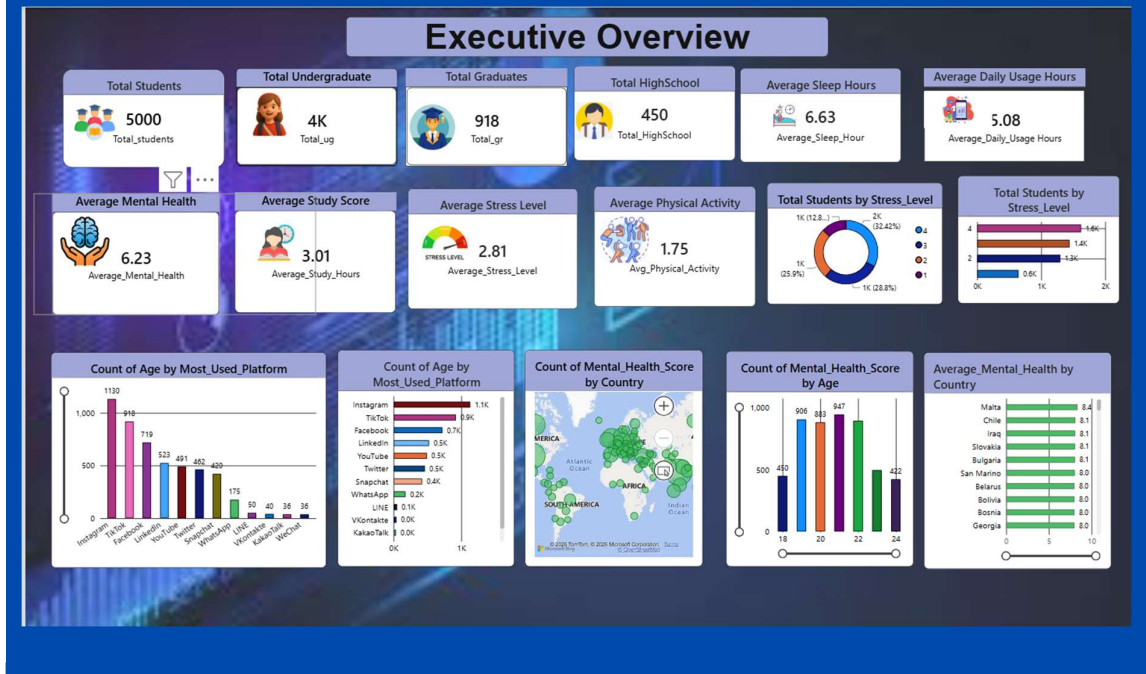
✓ Storytelling

✓ Business Recommendations

Dashboard Requirements

Participants must create a minimum of 5 Dashboard Pages.

1.Executive Overview



Purpose : Provide a high-level summary of student behavior and mental health indicators.

Required KPIs

- ✓ Total Students
- ✓ Average Mental Health Score
- ✓ Average Daily Usage Hours
- ✓ Average Study Hours
- ✓ Average Sleep Hours
- ✓ Average Stress Level

Suggested Visuals

- ✓ Mental Health Overview
- ✓ Platform Distribution
- ✓ Country Analysis
- ✓ Stress Analysis

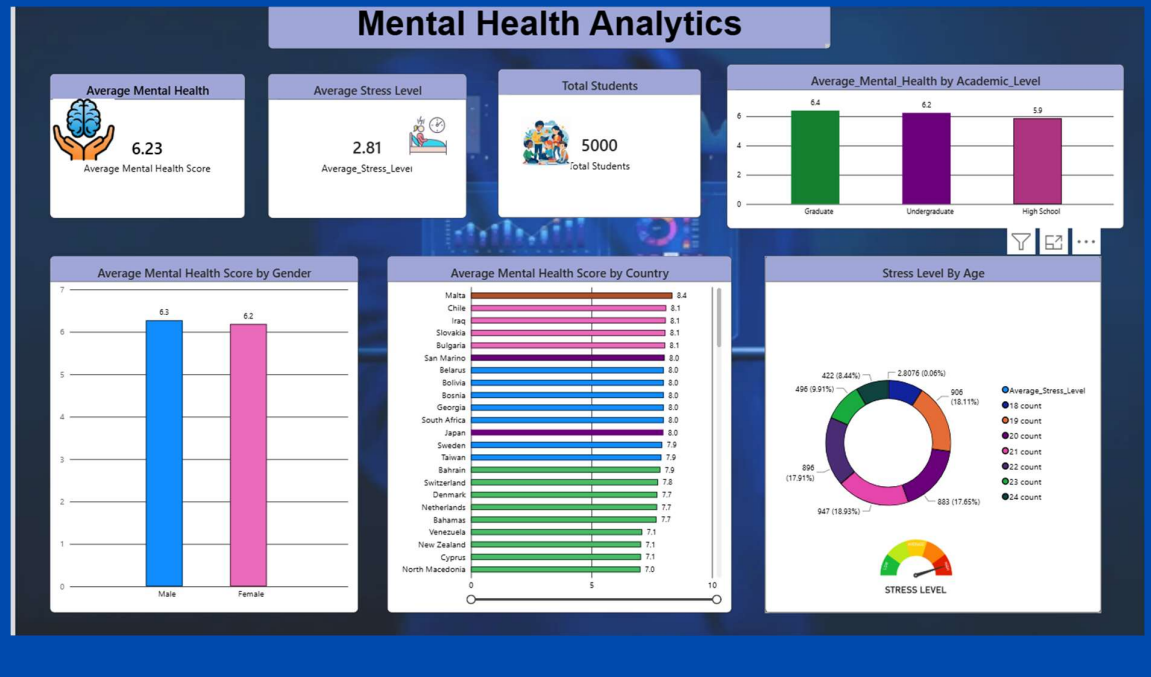
Social Media Analytics

Purpose : *Analyze social media usage patterns among students.*

Required Analysis

- ✓ *Most Used Platform*
- ✓ *Purpose of Usage*
- ✓ *Usage Hours Analysis*
- ✓ *Daily Unlock Analysis*
- ✓ *Platform Comparison*

3. Mental Health Analytics

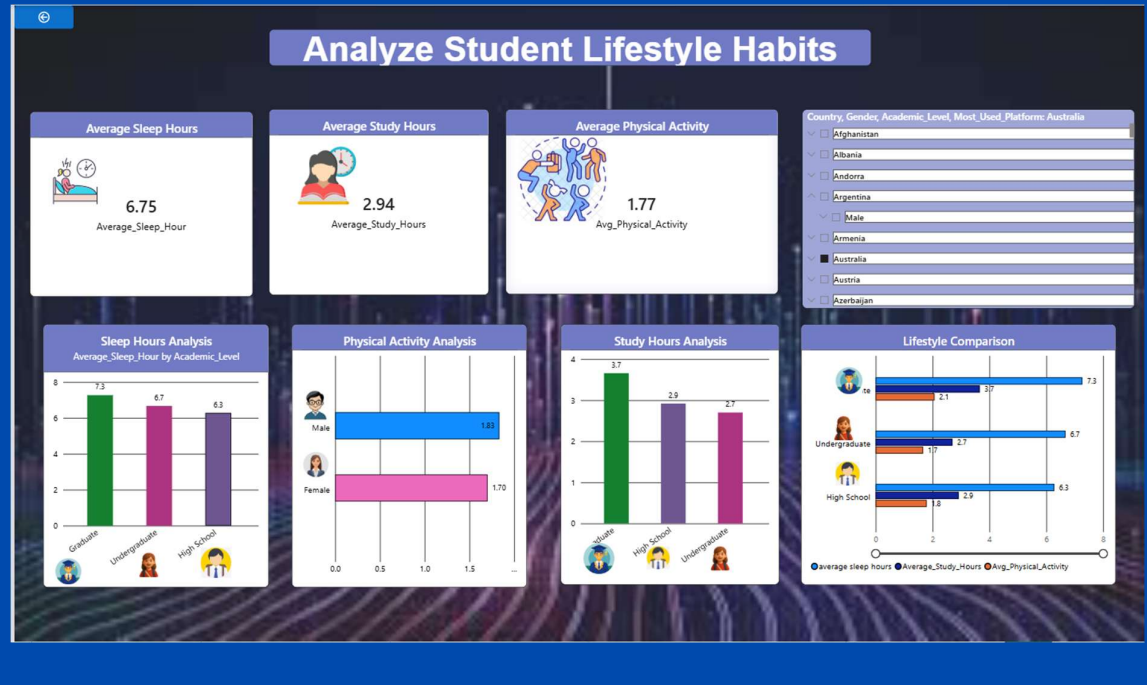


Purpose: Understand mental well-being and stress levels.

Required Analysis

- ✓ Mental Health Score by Gender
- ✓ Mental Health Score by Country
- ✓ Stress Level Distribution
- ✓ Mental Health Score by Academic Level

4.Lifestyle Analytics

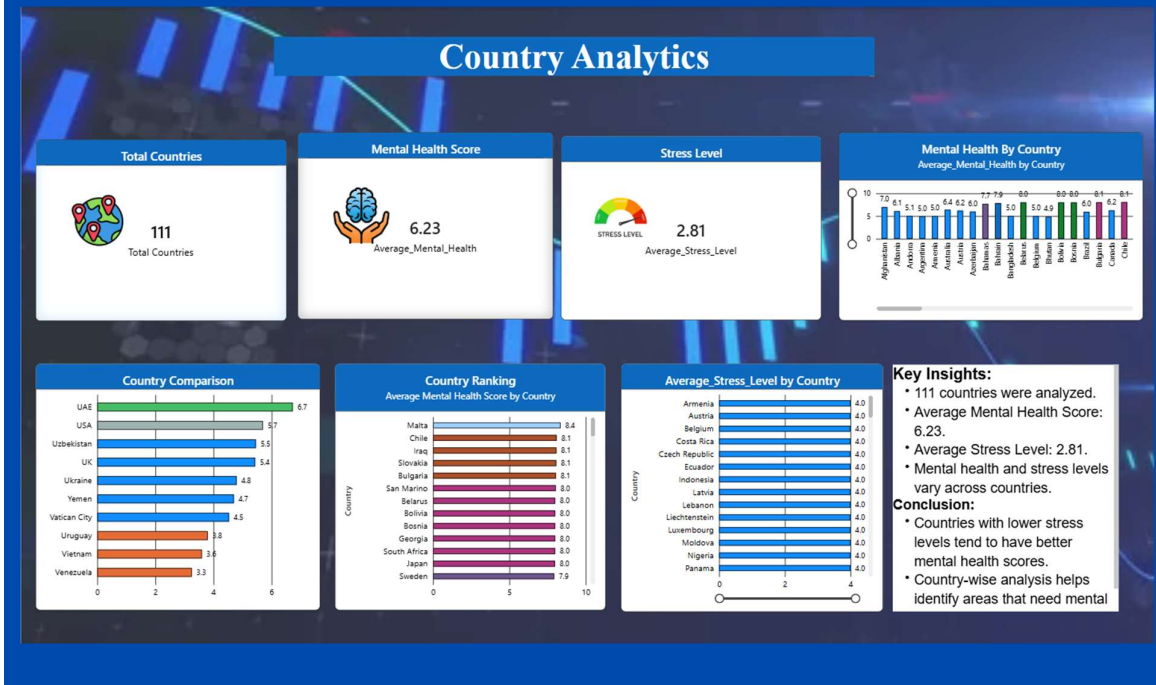


Purpose : *Analyze student lifestyle habits.*

Required Analysis

- ✓ Sleep Hours Analysis
- ✓ Physical Activity Analysis
- ✓ Study Hours Analysis
- ✓ Lifestyle Comparison

5. Country Analytics



Purpose : Compare student behavior across countries.

Required Analysis

✓ Country Comparison

✓ Country Ranking

✓ Mental Health by Country

✓ Stress Level by Country

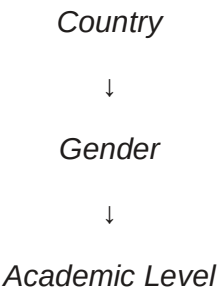
Mandatory Features

Purpose : Participants must implement the following concepts.

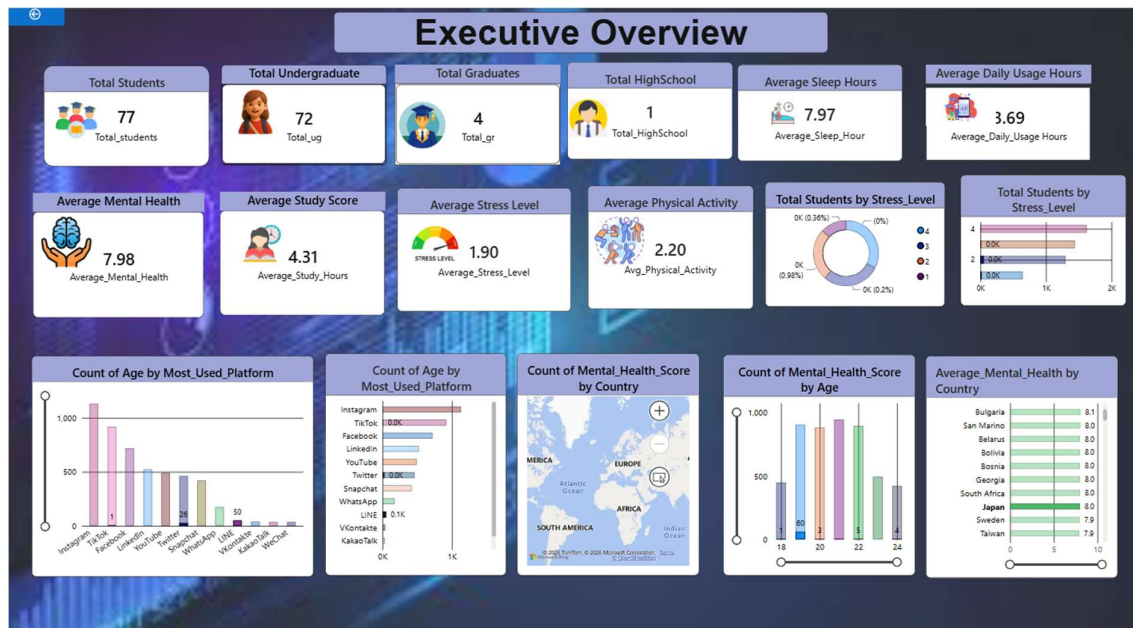
DRILL DOWN :



Examples :



DRILL THROUGH :



Examples

Country



Detailed Student Analysis

Platform



Platform Detail Analysis

CONDITIONAL FORMATTING :

Apply color logic using DAX.

Green = Healthy Score

Orange = Average Score

Red = Low Score

SLICERS :

Required Slicers : Country , Gender , Academic Level and Most Used Platform.

Storytelling

Example Story :

Students today spend a significant amount of time on social media, raising concerns about its impact on their mental health and academic performance. To understand this relationship, we analyzed data from students across 111 countries.

Our analysis revealed that students spend an average of 5.08 hours per day on social media, which is higher than their average study time of 3.01 hours per day. The average mental health score was 6.23/10, indicating moderate well-being among students.

We found that Instagram is the most popular platform, and High School students experience the highest stress levels. Additionally, students with better sleep patterns tend to have higher mental health scores and lower stress levels.

These insights highlight the importance of maintaining a healthy balance between social media usage, study time, and sleep to improve students' overall well-being and academic success.

Observation

Students spending more than 6 hours daily on social media show lower mental health scores.

Insight

Excessive screen time may impact well-being.

Project Deliverables

Participants must submit:

1. **PBIX FILE** : Submit the complete Power BI project containing all dashboard pages, visuals, KPIs, DAX measures, interactions, drill-down, drill-through and a fully functional dashboard solution.
2. **PROJECT DOCUMENTATION** : Prepare a project report covering : ✓ Project Objective ✓ Business Problem ✓ Dataset Overview ✓ KPIs Created ✓ Key Insights ✓ Business Recommendations
3. **DASHBOARD SCREENSHOTS** : Include screenshots of all major dashboard pages: ✓ Executive Overview ✓ Social Media Analytics ✓ Mental Health Analytics ✓ Lifestyle Analytics ✓ Country Analytics
4. **GITHUB REPOSITORY** : Create a GitHub repository and upload: ✓ PBIX File ✓ Documentation ✓ Dashboard Screenshots ✓ README File The README should clearly explain the project, objectives, dashboards, insights and recommendations.
5. **PROJECT PRESENTATION** : Prepare a professional presentation of 5–10 minutes demonstrating: ✓ Business Problem ✓ Dataset Overview ✓ Dashboard Walkthrough ✓ Key Insights ✓ Recommendations ✓ Project Conclusion

Submission Guidelines

Submission Format

Participants must submit all project deliverables in an organized and professional manner. The final submission should include the PBIX file, project documentation, dashboard screenshots, GitHub repository link, and presentation materials. All files must follow the prescribed naming conventions and folder structure to ensure easy review and evaluation. Participants are encouraged to upload their completed project to GitHub and showcase it on LinkedIn as part of their professional portfolio. Late,

incomplete, or improperly organized submissions may result in a reduction of evaluation marks.

Conclusion

“The dashboard should not only display charts.

It should answer meaningful questions.

It should generate insights.

It should support better decision making.”